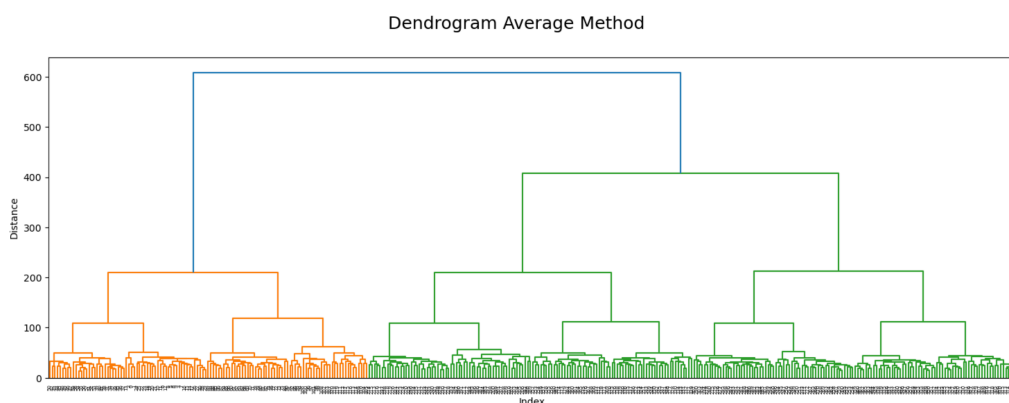
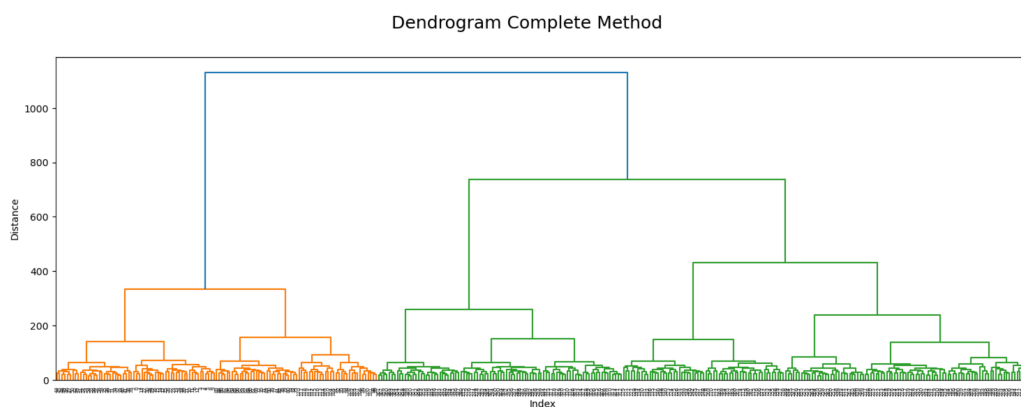
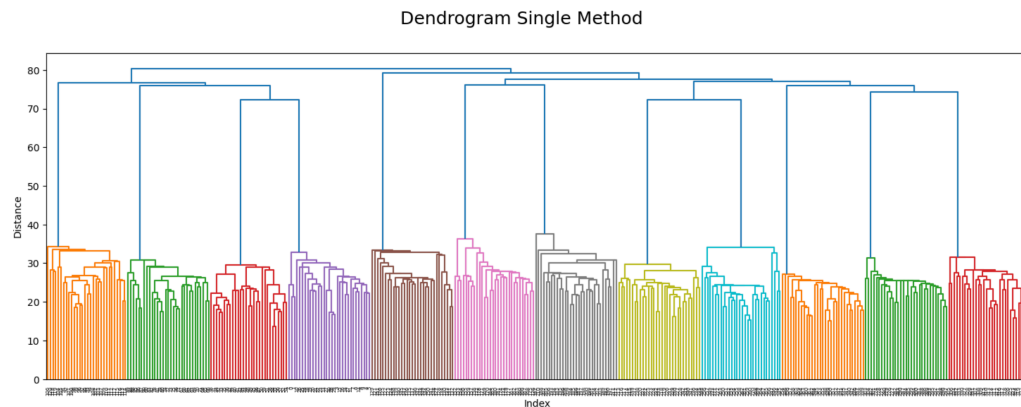


2.1 Unsupervised learning algorithms

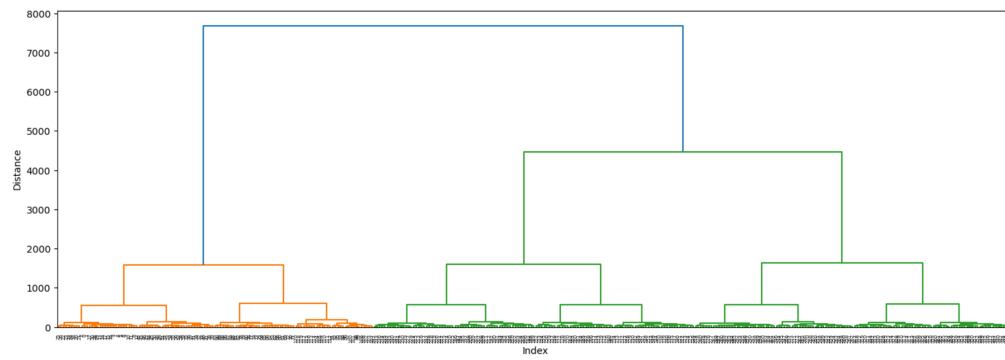
Create a dendrogram for the weather data using each of the four methods: single, complete, average, and ward. Record how many clusters you find using each method.

Basing off where the longest vertical lines are, it seems there are 12 for single method, 3 for complete, 3 for average, and 3 for the ward methods

Dendrograms – Unscaled Data

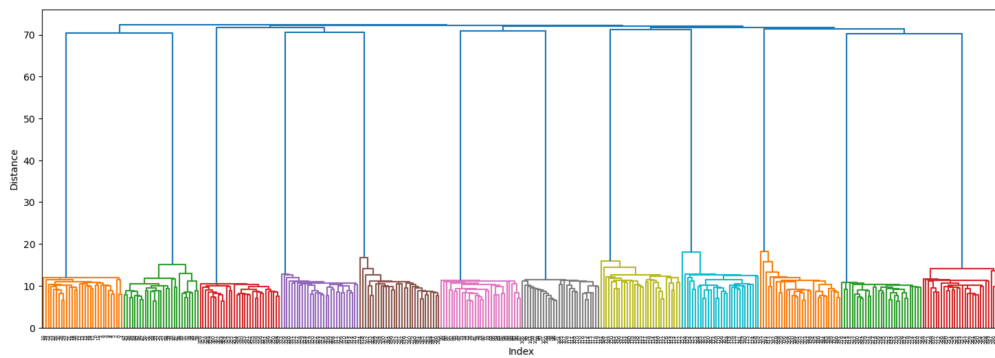


Dendrogram Ward Method

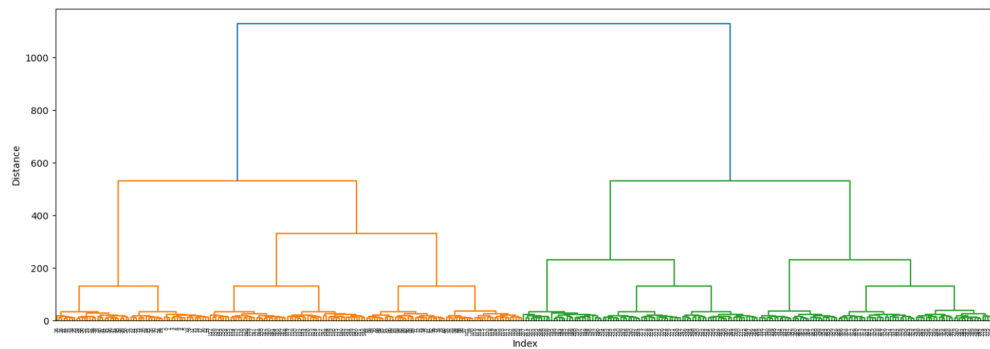


Run all four dendrograms with scaled data and record screenshots. (Note that the single method may cause the kernel to crash if the data is scaled. If this happens, run the single method with unscaled data.)

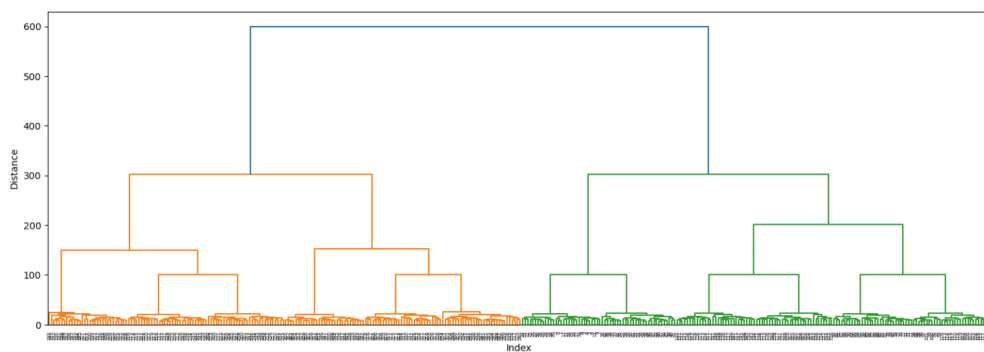
Dendrogram Single Method

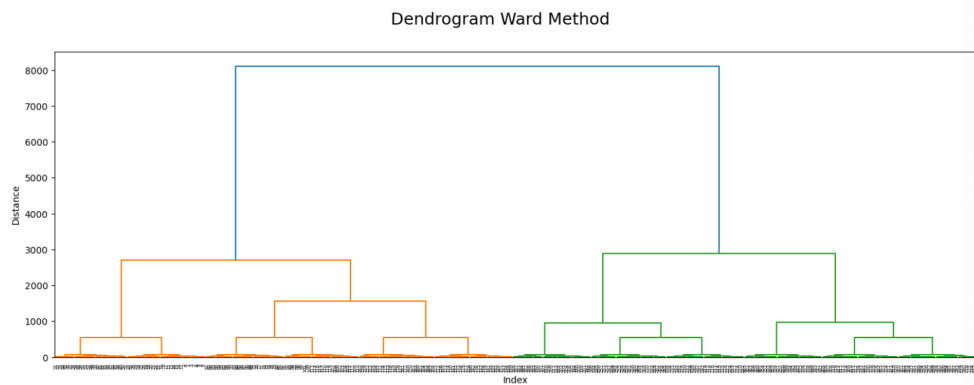


Dendrogram Complete Method



Dendrogram Average Method





Are there any relevant insights? Add them to your document together with the screenshots you took. Use screenshots of your graphs if needed.

Basing off where the longest vertical lines are, it seems there are still 12 for single method, but now 2 each for complete, average, and ward methods

This shows that scaling the data results in fewer clusters for the complete, average and ward methods, and the same 12 for single method.

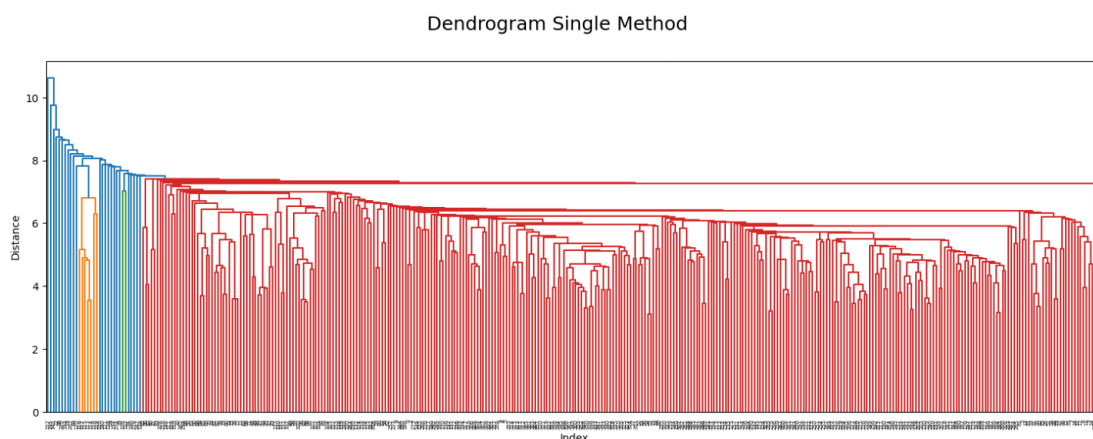
For scaled single, the average distance before the longest vertical line is around 10 compared to 35 for unscaled, showing that scaling the data significantly decreases the distance.

The complete, average, and ward methods have more equal clusters for scaled compared to unscaled. This would also be expected as scaling brings the values closer together and minimizes the impact of outliers in the unscaled data.

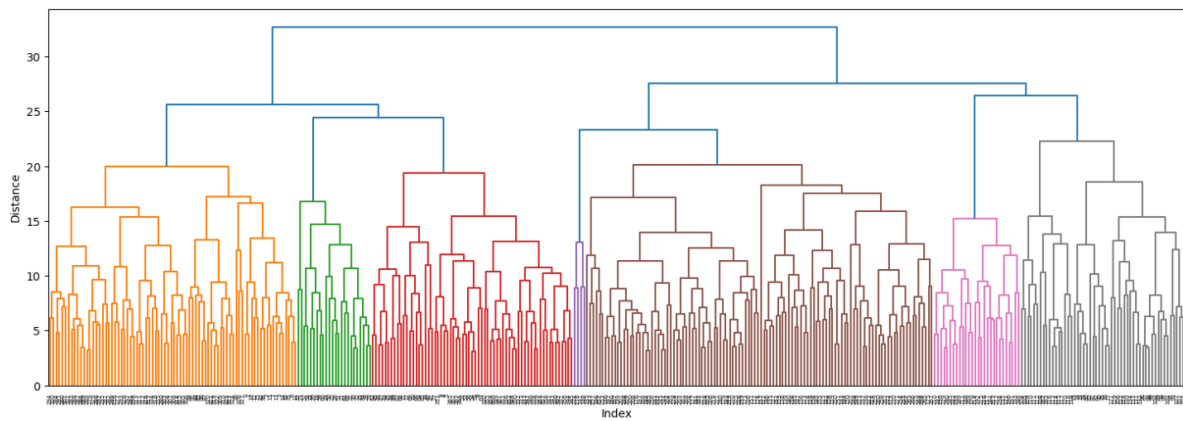
Is the clustering more conclusive? Do any of the clusters match the “pleasant weather” labels?

For all these dendrograms after apply PCA, the distance is greatly reduced. Having reduced to 18 components (one per station) the complete and average methods now have more clusters.

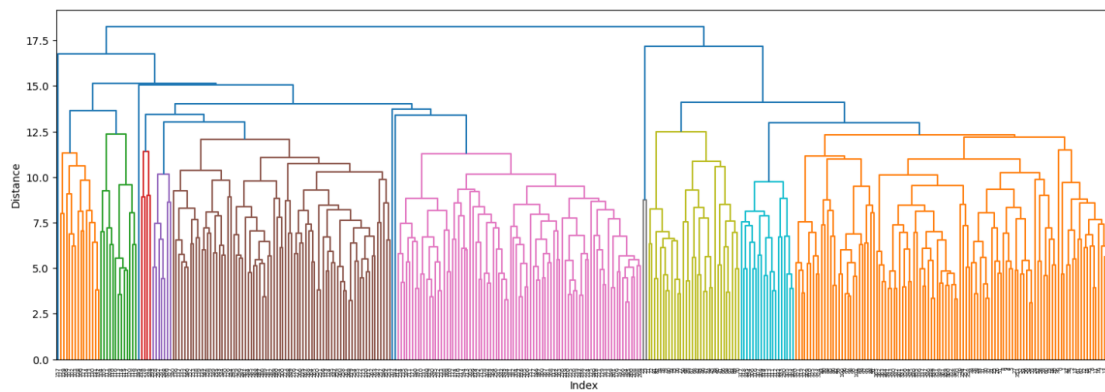
The single method has fewer clusters and the red cluster represents most of the data points, while the ward method is still 2 clusters and this could be seen to match the pleasant data which is mostly separated into 2 clusters with the more accurate KNN and ANN supervised models.



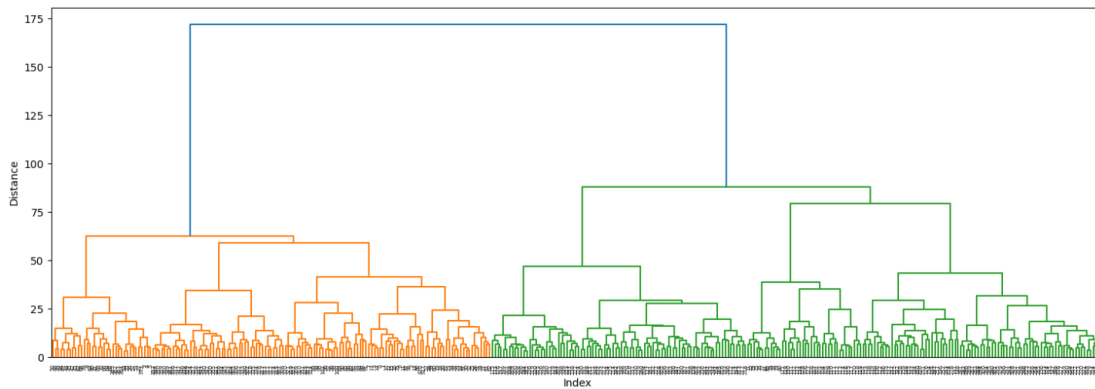
Dendrogram Complete Method



Dendrogram Average Method



Dendrogram Ward Method



Upload your Jupyter scripts here for your mentor to evaluate, along with your document (you may also upload a PDF) of observations on hierarchical clustering, any insights you found, your conclusion on whether scaled or unscaled data gave better results, the screenshots you took, and any other observations you want to share.

In conclusion, the scaled data works much better, as we would expect it, to reduce the distance and show clearer clusters. The PCA further reduced the distance for all the dendrograms. We would expect this having significantly reduced the data set to 18 components for just the 1984 data. However, we must consider the risk of removing key variables. I would like to do further analysis with a controlled reduction of certain variables, and this would help support my hypothesis focusing on wind speed.